MTCNN (Multi-task Cascaded Convolutional Neural Network) - detecția fețelor

este un instrument disponibil pentru detecția fețelor în imagini și secvențe video. Este utilizat frecvent în aplicații de prelucrare a imaginilor, sisteme de recunoaștere facială și aliniere facială. Pe lângă identificarea poziției feței printr-un dreptunghi de încadrare, MTCNN poate furniza și coordonatele unor puncte caracteristice ale feței (ochi, nas, colțurile gurii), ceea ce îl face util pentru etape ulterioare precum normalizarea sau recunoașterea.

Există implementări disponibile în Python, în special prin biblioteci bazate pe TensorFlow sau PyTorch, precum și integrarea în pachete orientate către recunoaștere facială (de exemplu în cadrul unor implementări FaceNet). Este considerat un detector clasic, stabil și relativ ușor de utilizat, potrivit atât pentru prototipare cât și pentru aplicații practice unde este necesară o detecție fiabilă a fețelor fără configurări complexe.

```
!pip install mtcnn
```

```
Collecting mtcnn
  Downloading mtcnn-1.0.0-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: joblib>=1.4.2 in
/usr/local/lib/python3.12/dist-packages (from mtcnn) (1.5.3)
Collecting lz4>=4.3.3 (from mtcnn)
  Downloading lz4-4.4.5-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.m
etadata (3.8 kB)
Downloading mtcnn-1.0.0-py3-none-any.whl (1.9 MB)
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anylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl
(1.4 MB)
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0:00:00
tcnn
Successfully installed lz4-4.4.5 mtcnn-1.0.0
```

```
import cv2
import matplotlib.pyplot as plt
from mtcnn import MTCNN
from google.colab import files
from IPython.display import display
from google.colab.patches import cv2_imshow
```

- OpenCV, folosit pentru citirea, prelucrarea și manipularea imaginilor (ex. încărcare fișiere, conversii, desenarea de dreptunghiuri pe imagine).
- Matplotlib, folosit pentru afișarea imaginilor și a rezultatelor în notebook

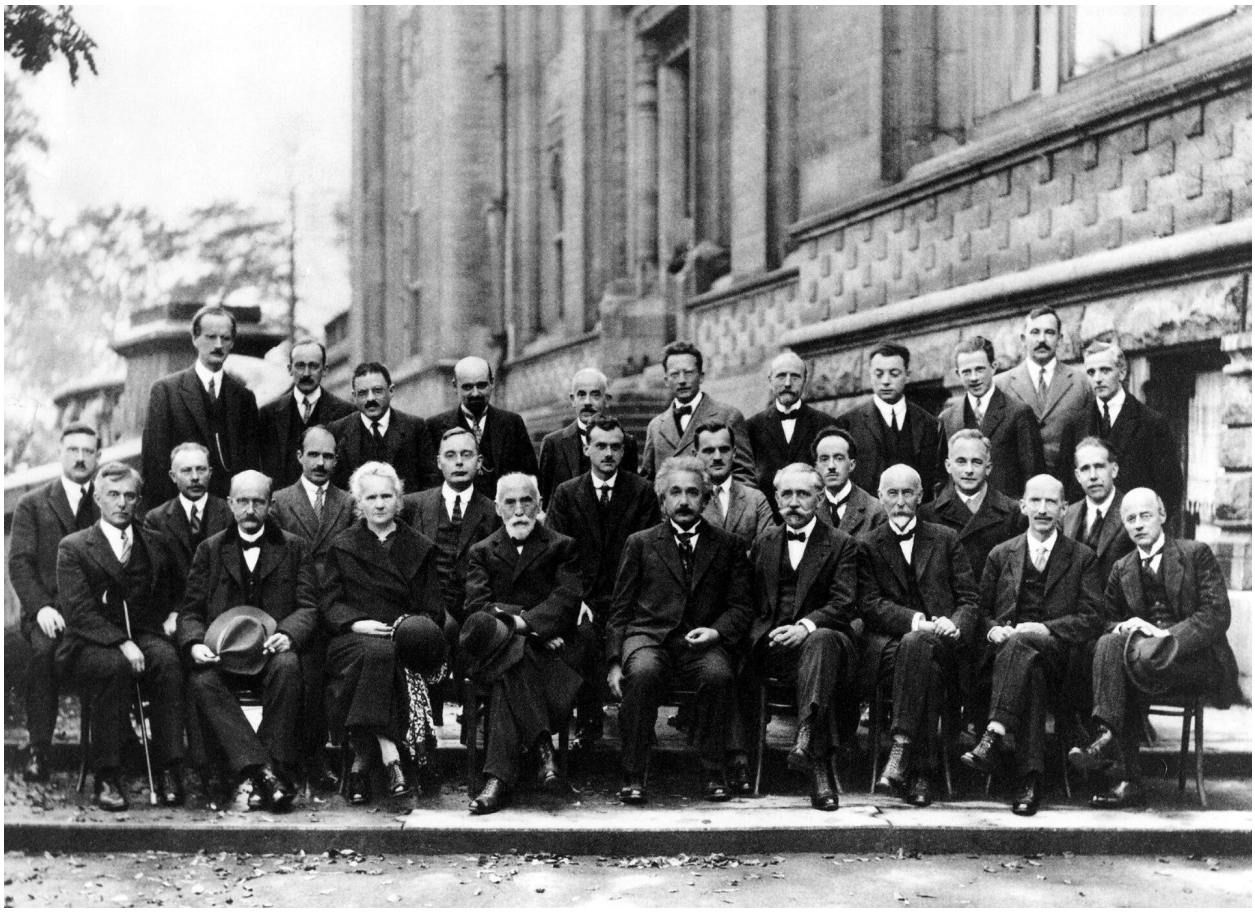
- clasa MTCNN, adică detectorul de fețe care poate fi rulat pe o imagine pentru a găsi fețele (și, de obicei, puncte faciale).
- permite încărcarea de fișiere din calculatorul local în sesiunea Colab
- util pentru a afișa obiecte în notebook
- funcție specială pentru Colab care afișează imagini OpenCV corect; cv2.imshow() nu funcționează în Colab, iar cv2_imshow() este alternativa recomandată.

```
detector = MTCNN()

print("Incarcati o imagine de test")
uploaded_test_image = files.upload()  # Allows user to upload a file
# Get the first uploaded file
uploaded_test_image_path = list(uploaded_test_image.keys())[0]
img = cv2.imread(uploaded_test_image_path)
cv2_imshow(img)

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

faces = detector.detect_faces(img_rgb)

image_copy = img.copy()
for face in faces:
    # Coordonatele casetei de detectie
    x, y, width, height = face['box']

    # Deseneaza casetele de detectie
    cv2.rectangle(image_copy, (x, y), (x + width, y + height), (0,
255, 0), 2)

    # Deseneaza punctele semnificative
    for landmark in face['keypoints'].values():
        cv2.circle(image_copy, landmark, 4, (0, 0, 255), -1)  # Puncte
rosii

cv2_imshow(image_copy)
```

```
Incarcati o imagine de test
```

```
<IPython.core.display.HTML object>
```

```
Saving Solvay_conference_1927.jpg to Solvay_conference_1927.jpg
```



YOLO (You Only Look Once) - detectia obiectelor

Ultralytics YOLO este o implementare modernă și ușor de utilizat a algoritmului YOLO (You Only Look Once) pentru detecția obiectelor în imagini și video. Biblioteca oferă un API simplu în Python, permite încărcarea rapidă a unor modele pre-antrenate și rularea detecției cu foarte puțin cod, fiind potrivită pentru experimente rapide în Google Colab sau în proiecte practice. Modelul procesează imaginea într-o singură trecere prin rețea și returnează coordonatele bounding box-urilor, scorurile de încredere și clasele obiectelor detectate.

```
!pip -q install -U ultralytics
```

```
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0:00:01 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 20.6 MB/s
eta 0:00:00
```
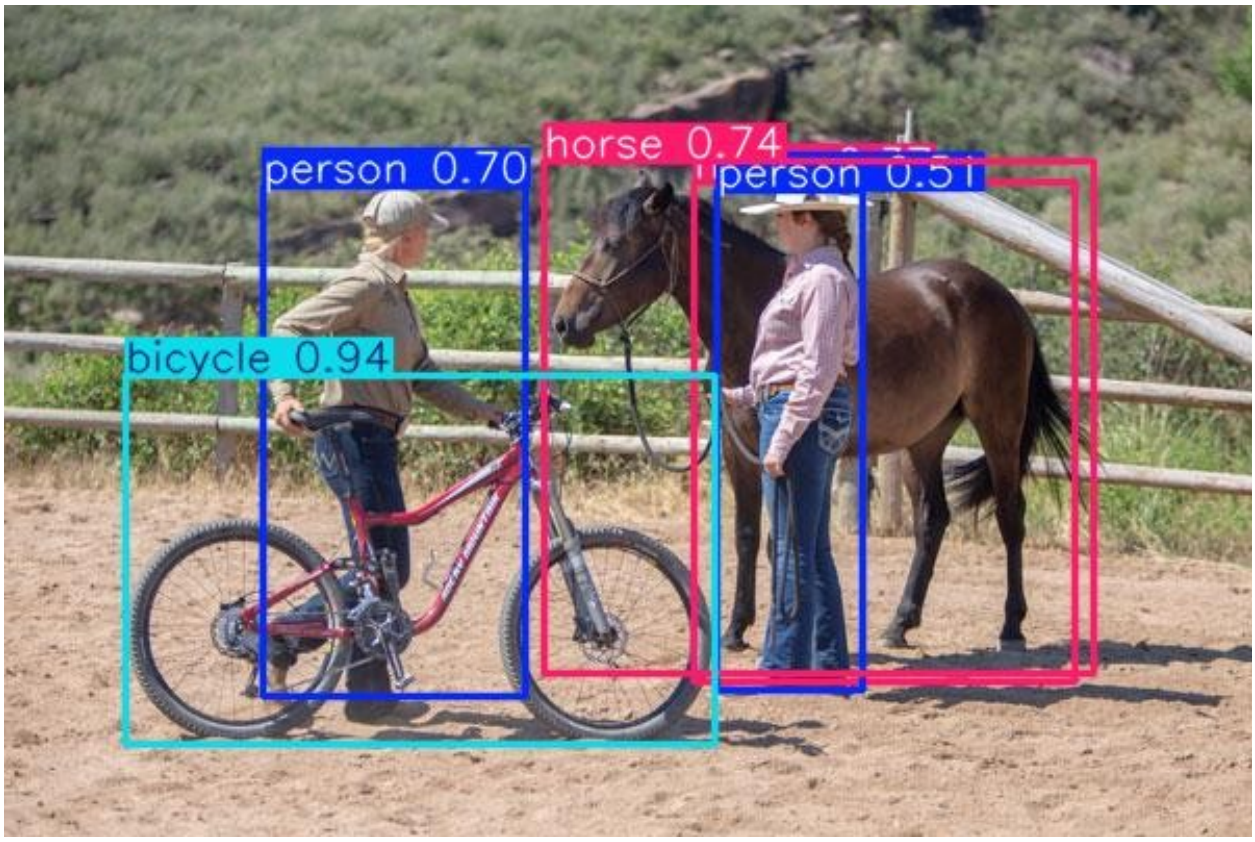

```
from ultralytics import YOLO
model = YOLO("yolov8n.pt")   # small + fast
```

```
Creating new Ultralytics Settings v0.0.6 file 
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to
'yolov8n.pt': 100% ━━━━━━━━━━━━━━━━ 6.2MB 69.3MB/s 0.1s
```

```python
print("Incarcati o imagine de test")
uploaded_test_image_2 = files.upload()  # Allows user to upload a file
# Get the first uploaded file
uploaded_test_image_path_2 = list(uploaded_test_image_2.keys())[0]

# Read the image as a numpy array
img2 = cv2.imread(uploaded_test_image_path_2)
results2 = model(img2)

annotated_2 = results2[0].plot()
cv2_imshow(annotated_2)
```

```
Incarcati o imagine de test

<IPython.core.display.HTML object>

Saving bike.jpg to bike.jpg

0: 448x640 2 persons, 1 bicycle, 2 horses, 402.6ms
Speed: 13.3ms preprocess, 402.6ms inference, 52.2ms postprocess per
image at shape (1, 3, 448, 640)
```



Detectron

Detectron este o bibliotecă open-source dezvoltată de Facebook AI Research (Meta) pentru detecția și segmentarea obiectelor în imagini.

```
!pip -q install
"git+https://github.com/facebookresearch/detectron2.git"

  Preparing metadata (setup.py) ...
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 50.2/50.2 kB 2.2 MB/s eta
0:00:00
etadata (setup.py) ... ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
88.9/88.9 kB 5.6 MB/s eta 0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 154.5/154.5 kB 11.0 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 36.7 MB/s eta
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0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 269.8/269.8 kB 17.3 MB/s eta
0:00:00
```

```python
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow

from detectron2.engine import DefaultPredictor
from detectron2.config import get_cfg
from detectron2 import model_zoo
from detectron2.utils.visualizer import Visualizer
from detectron2.data import MetadataCatalog
import matplotlib.pyplot as plt
import copy

cfg = get_cfg()
cfg.merge_from_file(model_zoo.get_config_file(
    "COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml"
))
cfg.MODEL.WEIGHTS = model_zoo.get_checkpoint_url(
    "COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml"
)
cfg.MODEL.ROI_HEADS.SCORE_THRESH_TEST = 0.5
cfg.MODEL.DEVICE = "cpu"
predictor = DefaultPredictor(cfg)
```

```
model_final_f10217.pkl: 178MB [00:01, 90.2MB/s]
```

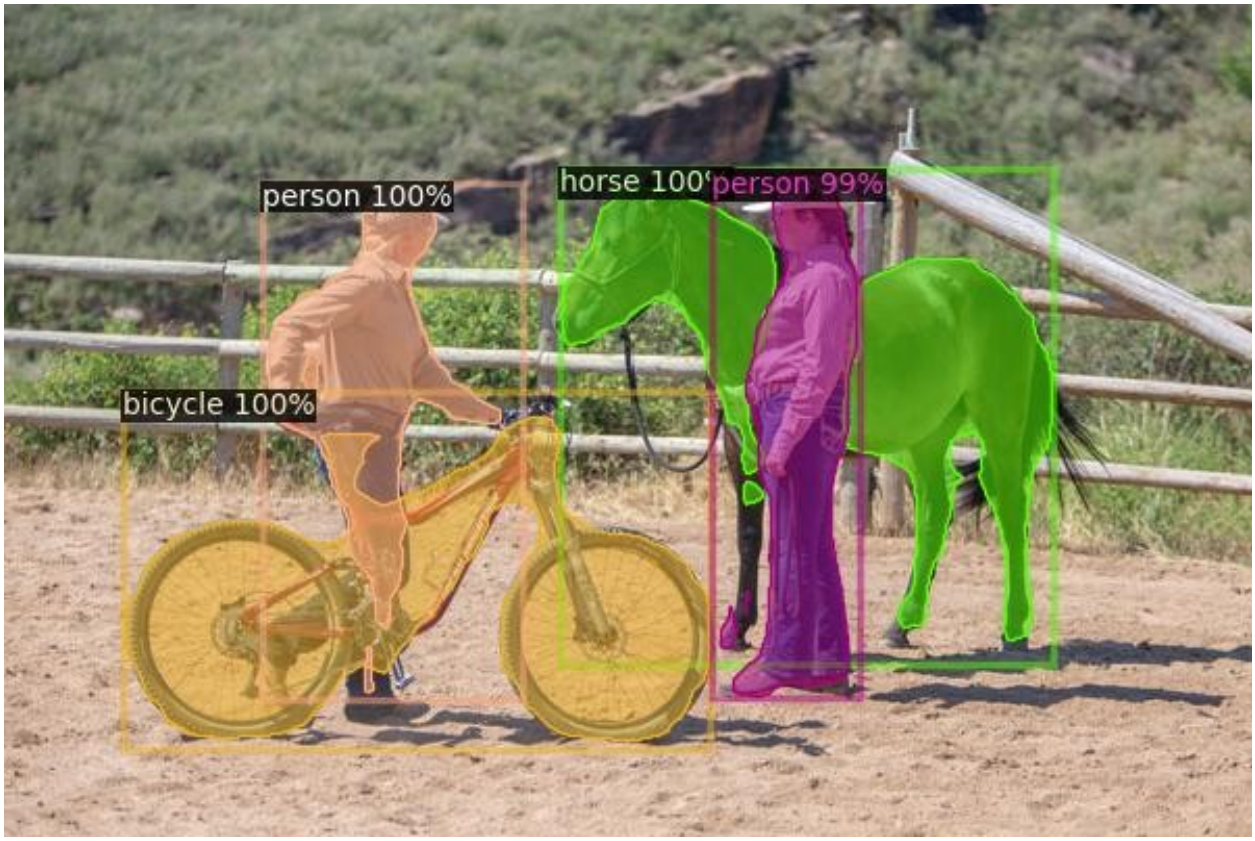
```python
uploaded_test_image_2 = files.upload()
uploaded_test_image_path_2 = list(uploaded_test_image_2.keys())[0]

img2 = cv2.imread(uploaded_test_image_path_2)
outputs2 = predictor(img2)

v = Visualizer(
    img2[:, :, ::-1],
    MetadataCatalog.get(cfg.DATASETS.TRAIN[0]),
    scale=1.0
)
vis_rgb_2 =
v.draw_instance_predictions(outputs2["instances"].to("cpu")).get_image
()

vis_bgr_2 = vis_rgb_2[:, :, ::-1]
cv2_imshow(vis_bgr_2)
```

```
<IPython.core.display.HTML object>
```

```
Saving bike.jpg to bike (1).jpg
```

```
/usr/local/lib/python3.12/dist-packages/torch/functional.py:505:
UserWarning: torch.meshgrid: in an upcoming release, it will be
required to pass the indexing argument. (Triggered internally at
/pytorch/aten/src/ATen/native/TensorShape.cpp:4381.)
```

```
  return _VF.meshgrid(tensors, **kwargs)  # type: ignore[attr-defined]
W0301 18:35:49.591000 203 torch/fx/_symbolic_trace.py:53]
is_fx_tracing will return true for both fx.symbolic_trace and
torch.export. Please use is_fx_tracing_symbolic_tracing() for
specifically fx.symbolic_trace or torch.compiler.is_compiling() for
specifically torch.export/compile.
```



Deoldify

```
!git clone -q https://github.com/jantic/DeOldify.git /content/DeOldify
%cd /content/DeOldify
!pip -q install -r requirements-colab.txt

!mkdir -p models
!wget -q -O models/ColorizeArtistic_gen.pth
https://data.deepai.org/deoldify/ColorizeArtistic_gen.pth
!wget -q -O models/ColorizeStable_gen.pth
https://data.deepai.org/deoldify/ColorizeStable_gen.pth


/content/DeOldify
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0:00:00
```

```
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0:00:00
etadata (setup.py) ... etadata (setup.py) ...
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0:00:00
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0:00:00
gaug (setup.py) ... l-py3 (setup.py) ...
```

```
import functools
import torch
from PIL import Image, ImageEnhance


_torch_load = torch.load
def torch_load_compat(*args, **kwargs):
    kwargs.setdefault("weights_only", False)
    return _torch_load(*args, **kwargs)

torch.load = torch_load_compat

from deoldify.visualize import get_image_colorizer
colorizer = get_image_colorizer(artistic=True)
```

```
/content/DeOldify/deoldify/visualize.py:223: SyntaxWarning: invalid
escape sequence '\.'
  if re.search('.*?\.jpg', f):
/content/DeOldify/fastai/gen_doc/gen_notebooks.py:63: SyntaxWarning:
invalid escape sequence '\s'
  match = re.match(f"^({key})\s*=\s*.*", codestr)
/content/DeOldify/fastai/gen_doc/gen_notebooks.py:216: SyntaxWarning:
invalid escape sequence '\('
  if re.search(f"update_nb_metadata\('{fn}'", c['source']): return c
/content/DeOldify/fastai/gen_doc/nbdoc.py:26: SyntaxWarning: invalid
escape sequence '\*'
  arg_prefixes = {inspect._VAR_POSITIONAL: '\*',
inspect._VAR_KEYWORD:'\*\*'}
/content/DeOldify/fastai/gen_doc/nbdoc.py:26: SyntaxWarning: invalid
escape sequence '\*'
  arg_prefixes = {inspect._VAR_POSITIONAL: '\*',
inspect._VAR_KEYWORD:'\*\*'}
/content/DeOldify/fastai/gen_doc/nbdoc.py:56: SyntaxWarning: invalid
escape sequence '\['
  return f'`Optional`\[{type_repr(args[0])}\]'
/content/DeOldify/fastai/gen_doc/nbdoc.py:56: SyntaxWarning: invalid
escape sequence '\]'
  return f'`Optional`\[{type_repr(args[0])}\]'
/content/DeOldify/fastai/gen_doc/nbdoc.py:58: SyntaxWarning: invalid
```

```
escape sequence '\['
  return f'{link_type(t)}\[{reprs}\]'
/content/DeOldify/fastai/gen_doc/nbdoc.py:58: SyntaxWarning: invalid
escape sequence '\]'
  return f'{link_type(t)}\[{reprs}\]'
/content/DeOldify/fastai/gen_doc/nbdoc.py:172: SyntaxWarning: invalid
escape sequence '\['
  BT_REGEX = re.compile("\[`([^`]*)`\](?:\([^)]*\))|`([^`]*)`") #
matches [`key`](link) or `key`
/content/DeOldify/fastai/gen_doc/docstrings.py:26: SyntaxWarning:
invalid escape sequence '\*'
  NEW_REGEX = re.compile(f":(?P<field>{FIELDS}) (?P<name>[\*\w]+): (?
P<doc>.*?)"
/content/DeOldify/fastai/gen_doc/docstrings.py:27: SyntaxWarning:
invalid escape sequence '\Z'
  f"(?:(?=:(?:{FIELDS}|return|raises))|\Z)", re.S)
/content/DeOldify/fastai/gen_doc/nbtest.py:111: SyntaxWarning: invalid
escape sequence '\w'
  exact_re = '' if exact_match else '\w*'
/content/DeOldify/fastai/gen_doc/nbtest.py:112: SyntaxWarning: invalid
escape sequence '\w'
  return re.match(f'test_{subdir}\w*{Path(fp).stem}{exact_re}\.py',
file_name)
/content/DeOldify/fastai/gen_doc/nbtest.py:112: SyntaxWarning: invalid
escape sequence '\.'
  return re.match(f'test_{subdir}\w*{Path(fp).stem}{exact_re}\.py',
file_name)
/content/DeOldify/fastai/gen_doc/nbtest.py:155: SyntaxWarning: invalid
escape sequence '\s'
  if re.match(f'.*[\s\.\(]{fn_name}[\.\(]', line):
/content/DeOldify/fastai/gen_doc/nbtest.py:155: SyntaxWarning: invalid
escape sequence '\.'
  if re.match(f'.*[\s\.\(]{fn_name}[\.\(]', line):
/content/DeOldify/fastai/gen_doc/nbtest.py:164: SyntaxWarning: invalid
escape sequence '\s'
  test_name = re.match(f'\s*def (test_\w*)', line_text).groups(0)[0]
/content/DeOldify/fastai/gen_doc/doctest.py:121: SyntaxWarning:
invalid escape sequence '\s'
  if re.match(f'\s*def test', l):  return (lineno - idx), l # 1 based
index for github
/content/DeOldify/fastai/gen_doc/doctest.py:122: SyntaxWarning:
invalid escape sequence '\w'
  if re.match(f'\w+', l):  break # top level indent - out of function
scope
/content/DeOldify/fastai/data_block.py:451: UserWarning: Your training
set is empty. If this is by design, pass `ignore_empty=True` to remove
this warning.
  warn("Your training set is empty. If this is by design, pass
`ignore_empty=True` to remove this warning.")
```

```
/content/DeOldify/fastai/data_block.py:453: UserWarning: Your
validation set is empty. If this is by design, use `split_none()`
                 or pass `ignore_empty=True` when labelling to remove
this warning.
  warn("""Your validation set is empty. If this is by design, use
`split_none()`
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
:424: UserWarning: This DataLoader will create 8 worker processes in
total. Our suggested max number of worker in current system is 2,
which is smaller than what this DataLoader is going to create. Please
be aware that excessive worker creation might get DataLoader running
slow or even freeze, lower the worker number to avoid potential
slowness/freeze if necessary.
  self.check_worker_number_rationality()
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:2
08: UserWarning: The parameter 'pretrained' is deprecated since 0.13
and may be removed in the future, please use 'weights' instead.
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:2
23: UserWarning: Arguments other than a weight enum or `None` for
'weights' are deprecated since 0.13 and may be removed in the future.
The current behavior is equivalent to passing
`weights=ResNet34_Weights.IMAGENET1K_V1`. You can also use
`weights=ResNet34_Weights.DEFAULT` to get the most up-to-date weights.
  warnings.warn(msg)

Downloading: "https://download.pytorch.org/models/resnet34-
b627a593.pth" to /root/.cache/torch/hub/checkpoints/resnet34-
b627a593.pth

100%|██████████| 83.3M/83.3M [00:01<00:00, 45.6MB/s]
/usr/local/lib/python3.12/dist-packages/torch/nn/utils/weight_norm.py:
144: FutureWarning: `torch.nn.utils.weight_norm` is deprecated in
favor of `torch.nn.utils.parametrizations.weight_norm`.
  WeightNorm.apply(module, name, dim)
```

```python
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow

uploaded_test_image = files.upload()
uploaded_test_image_path = list(uploaded_test_image.keys())[0]

img = cv2.imread(uploaded_test_image_path)
cv2_imshow(img)

result_path = colorizer.plot_transformed_image(
    path=uploaded_test_image_path,
    render_factor=55,
```

```
    compare=False
)
```

```
<IPython.core.display.HTML object>
```

```
Saving person-black-and-white-group-people-old-monochrome-black-white-family-photograph-historical-monochrome-photography-social-group-1358794.jpg to person-black-and-white-group-people-old-monochrome-black-white-family-photograph-historical-monochrome-photography-social-group-1358794.jpg
```

Background remover

```
!pip -q install "rembg[cpu]" --upgrade
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 16.6/16.6 MB 87.6 MB/s eta
0:00:00
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0:00:00
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0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 54.9/54.9 kB 3.9 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 13.6/13.6 MB 43.1 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.7/43.7 kB 335.1 kB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 76.7 MB/s eta
```

```
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 56.3/56.3 MB 14.1 MB/s eta
0:00:00
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
gradio 5.50.0 requires pillow<12.0,>=8.0, but you have pillow 12.1.1
which is incompatible.
tensorflow 2.19.0 requires numpy<2.2.0,>=1.26.0, but you have numpy
2.4.2 which is incompatible.
```

```
!pip -q install --upgrade --force-reinstall "numpy==2.0.2"
"pillow==11.3.0"
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.9/60.9 kB 2.5 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 19.2/19.2 MB 78.3 MB/s eta
0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.6/6.6 MB 104.2 MB/s eta
0:00:00
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
rembg 2.0.72 requires numpy<3.0.0,>=2.3.0, but you have numpy 2.0.2
which is incompatible.
rembg 2.0.72 requires pillow<13.0.0,>=12.1.0, but you have pillow
11.3.0 which is incompatible.
```

```
from google.colab import files
from PIL import Image
from rembg import remove, new_session
import io
import cv2
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt
import numpy as np

uploaded2 = files.upload()
uploaded2_path = list(uploaded2.keys())[0]
img2 = cv2.imread(uploaded2_path)

session2 = new_session("u2net")
out2 = remove(img2, session=session2)

a = out2[:, :, 3]                  # alpha channel
ys, xs = np.where(a > 0)           # or a > 10 to ignore faint edges

if len(xs) == 0 or len(ys) == 0:
    out2_cropped = out2            # nothing detected
else:
```

```
    x0, x1 = xs.min(), xs.max() + 1
    y0, y1 = ys.min(), ys.max() + 1
    out2_cropped = out2[y0:y1, x0:x1]


plt.figure(figsize=(16, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img2, cv2.COLOR_BGR2RGB))
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(out2_cropped, cv2.COLOR_BGRA2RGBA))
plt.axis("off")

plt.show()
```

```
<IPython.core.display.HTML object>
```

```
Downloading data from
'https://github.com/danielgatis/rembg/releases/download/v0.0.0/u2net.o
nnx' to file '/root/.u2net/u2net.onnx'.
```

```
Saving car.png to car.png
```


```
{"model_id":"c61b36ee09cd456eb04a059d19eba9fc","version_major":2,"vers
ion_minor":0}
```

Descrierea imaginilor

```
!pip install -U transformers torch
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.2.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
```

```
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.4.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch) (3.24.3)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.3.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: shellingham in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: typer>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from typer-slim-
```

```
>transformers) (0.24.1)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (2026.1.4)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0-
>transformers) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer>=0.24.0->typer-slim->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer>=0.24.0->typer-slim->transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (0.1.2)
Downloading transformers-5.2.0-py3-none-any.whl (10.4 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.4/10.4 MB 51.0 MB/s eta
0:00:00
ers
  Attempting uninstall: transformers
    Found existing installation: transformers 5.0.0
    Uninstalling transformers-5.0.0:
      Successfully uninstalled transformers-5.0.0
Successfully installed transformers-5.2.0
```

```python
import torch
from google.colab import files
from transformers import AutoConfig, BlipForConditionalGeneration,
BlipProcessor
from google.colab.patches import cv2_imshow
import cv2
```

```
model_id = "Salesforce/blip-image-captioning-base"
config = AutoConfig.from_pretrained(model_id)
config.tie_word_embeddings = False
processor = BlipProcessor.from_pretrained(model_id)

device = "cuda" if torch.cuda.is_available() else "cpu"
model = BlipForConditionalGeneration.from_pretrained(model_id,
config=config).to(device)
model.eval()
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(
INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary
Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary
Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-
cache/models/Salesforce/blip-image-captioning-base/
82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json
"HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-
cache/models/Salesforce/blip-image-captioning-base/
82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

HTTP Request: GET
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json
"HTTP/1.1 200 OK"

{"model_id":"76ef773bf1014c1d8167b8eaa139a026","version_major":2,"vers
ion_minor":0}

INFO:httpx:HTTP Request: GET
https://huggingface.co/api/models/Salesforce/blip-image-captioning-
```

```
base/tree/main/additional_chat_templates?recursive=false&expand=false
"HTTP/1.1 404 Not Found"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-
image-captioning-base/tree/main/additional_chat_templates?
recursive=false&expand=false "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/processor_config.json "HTTP/1.1 404
Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/processor_config.json "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/chat_template.json "HTTP/1.1 404
Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/chat_template.json "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/chat_template.jinja "HTTP/1.1 404
Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/chat_template.jinja "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/audio_tokenizer_config.json
"HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/audio_tokenizer_config.json "HTTP/1.1 404
Not Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/processor_config.json "HTTP/1.1 404
Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/processor_config.json "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/preprocessor_config.json "HTTP/1.1
307 Temporary Redirect"
```

```
HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/preprocessor_config.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

{"model_id":"63a75b8b869d4bdaac40783c50ad00a2","version_major":2,"version_minor":0}

The image processor of type `BlipImageProcessor` is now loaded as a fast processor by default, even if the model checkpoint was saved with a slow processor. This is a breaking change and may produce slightly different outputs. To continue using the slow processor, instantiate this class with `use_fast=False`.
INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/processor_config.json "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/processor_config.json "HTTP/1.1 404 Not Found"

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
```

```
INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/preprocessor_config.json "HTTP/1.1 307 Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/preprocessor_config.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/preprocessor_config.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/tokenizer_config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/tokenizer_config.json "HTTP/1.1 200 OK"
```

```
INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/tokenizer_config.json "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/tokenizer_config.json "HTTP/1.1 200 OK"

{"model_id":"5b377e5922764891bb9057fd4ea04f3b","version_major":2,"version_minor":0}

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/tree/main/additional_chat_templates?recursive=false&expand=false "HTTP/1.1 404 Not Found"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/tree/main/additional_chat_templates?recursive=false&expand=false "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/tree/main?recursive=true&expand=false "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/tree/main?recursive=true&expand=false "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/vocab.txt "HTTP/1.1 307 Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/vocab.txt "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/vocab.txt "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/vocab.txt "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/vocab.txt "HTTP/1.1 200 OK"
```

```
HTTP Request: GET
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/vocab.txt
"HTTP/1.1 200 OK"

{"model_id":"bfb784f00ab54308a3bab95d751d372d","version_major":2,"vers
ion_minor":0}

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/tokenizer.json "HTTP/1.1 307
Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/tokenizer.json "HTTP/1.1 307 Temporary
Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-
cache/models/Salesforce/blip-image-captioning-base/
82a37760796d32b1411fe092ab5d4e227313294b/tokenizer.json "HTTP/1.1 200
OK"

HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/
tokenizer.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-
cache/models/Salesforce/blip-image-captioning-base/
82a37760796d32b1411fe092ab5d4e227313294b/tokenizer.json "HTTP/1.1 200
OK"

HTTP Request: GET
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/
tokenizer.json "HTTP/1.1 200 OK"

{"model_id":"1f1af4b4842343edb49bd88828e2206c","version_major":2,"vers
ion_minor":0}

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/added_tokens.json "HTTP/1.1 404 Not
Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/added_tokens.json "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/special_tokens_map.json "HTTP/1.1
307 Temporary Redirect"
```

```
HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/special_tokens_map.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/special_tokens_map.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/special_tokens_map.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/special_tokens_map.json "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/special_tokens_map.json "HTTP/1.1 200 OK"

{"model_id":"e8d134ea3029493383eae9a93b601ee9","version_major":2,"version_minor":0}

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/adapter_config.json "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/adapter_config.json "HTTP/1.1 404 Not Found"
```

```
INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors.index.json "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors.index.json "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/pytorch_model.bin "HTTP/1.1 302 Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/pytorch_model.bin "HTTP/1.1 302 Found"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/xet-read-token/82a37760796d32b1411fe092ab5d4e227313294b "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/xet-read-token/82a37760796d32b1411fe092ab5d4e227313294b "HTTP/1.1 200 OK"

{"model_id":"f63f650ee11a49038cb203cc00a24cb1","version_major":2,"version_minor":0}

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/main/model.safetensors "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base "HTTP/1.1 200 OK"
```

```
INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/commits/main "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/commits/main "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/discussions?p=0 "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/discussions?p=0 "HTTP/1.1 200 OK"

{"model_id":"8685cc90ae68465db54ffec75bc9cf21","version_major":2,"version_minor":0}

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/commits/refs%2Fpr%2F52 "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/commits/refs%2Fpr%2F52 "HTTP/1.1 200 OK"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/refs%2Fpr%2F52/model.safetensors.index.json "HTTP/1.1 404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/refs%2Fpr%2F52/model.safetensors.index.json "HTTP/1.1 404 Not Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/refs%2Fpr%2F52/model.safetensors "HTTP/1.1 302 Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-captioning-base/resolve/refs%2Fpr%2F52/model.safetensors "HTTP/1.1 302 Found"

INFO:httpx:HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/xet-read-token/4c26dfece70e02028433dd192458a54b390b85d2 "HTTP/1.1 200 OK"

HTTP Request: GET https://huggingface.co/api/models/Salesforce/blip-image-captioning-base/xet-read-token/4c26dfece70e02028433dd192458a54b390b85d2 "HTTP/1.1 200 OK"

{"model_id":"fcebc6f0e01149448a4eb50ce1dc225a","version_major":2,"version_minor":0}
```

```
The tied weights mapping and config for this model specifies to tie
text_decoder.cls.predictions.bias to
text_decoder.cls.predictions.decoder.bias, but both are present in the
checkpoints, so we will NOT tie them. You should update the config
with `tie_word_embeddings=False` to silence this warning
The tied weights mapping and config for this model specifies to tie
text_decoder.bert.embeddings.word_embeddings.weight to
text_decoder.cls.predictions.decoder.weight, but both are present in
the checkpoints, so we will NOT tie them. You should update the config
with `tie_word_embeddings=False` to silence this warning
BlipForConditionalGeneration LOAD REPORT from: Salesforce/blip-image-
captioning-base
Key                                          | Status     |  |
---------------------------------------------+------------+--+-
text_decoder.bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/generation_config.json "HTTP/1.1
404 Not Found"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/generation_config.json "HTTP/1.1 404 Not
Found"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/Salesforce/blip-
image-captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary
Redirect"

HTTP Request: HEAD https://huggingface.co/Salesforce/blip-image-
captioning-base/resolve/main/config.json "HTTP/1.1 307 Temporary
Redirect"

INFO:httpx:HTTP Request: HEAD https://huggingface.co/api/resolve-
cache/models/Salesforce/blip-image-captioning-base/
82a37760796d32b1411fe092ab5d4e227313294b/config.json "HTTP/1.1 200 OK"

HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/Salesforce/blip-image-
captioning-base/82a37760796d32b1411fe092ab5d4e227313294b/config.json
"HTTP/1.1 200 OK"

BlipForConditionalGeneration(
  (vision_model): BlipVisionModel(
    (embeddings): BlipVisionEmbeddings(
      (patch_embedding): Conv2d(3, 768, kernel_size=(16, 16),
stride=(16, 16))
    )
    (encoder): BlipEncoder(
```

```
      (layers): ModuleList(
        (0-11): 12 x BlipEncoderLayer(
          (self_attn): BlipAttention(
            (dropout): Dropout(p=0.0, inplace=False)
            (qkv): Linear(in_features=768, out_features=2304,
bias=True)
            (projection): Linear(in_features=768, out_features=768,
bias=True)
          )
          (layer_norm1): LayerNorm((768,), eps=1e-05,
elementwise_affine=True)
          (mlp): BlipMLP(
            (activation_fn): GELUActivation()
            (fc1): Linear(in_features=768, out_features=3072,
bias=True)
            (fc2): Linear(in_features=3072, out_features=768,
bias=True)
          )
          (layer_norm2): LayerNorm((768,), eps=1e-05,
elementwise_affine=True)
        )
      )
    )
    (post_layernorm): LayerNorm((768,), eps=1e-05,
elementwise_affine=True)
  )
  (text_decoder): BlipTextLMHeadModel(
    (bert): BlipTextModel(
      (embeddings): BlipTextEmbeddings(
        (word_embeddings): Embedding(30524, 768, padding_idx=0)
        (position_embeddings): Embedding(512, 768)
        (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
        (dropout): Dropout(p=0.0, inplace=False)
      )
      (encoder): BlipTextEncoder(
        (layer): ModuleList(
          (0-11): 12 x BlipTextLayer(
            (attention): BlipTextAttention(
              (self): BlipTextSelfAttention(
                (query): Linear(in_features=768, out_features=768,
bias=True)
                (key): Linear(in_features=768, out_features=768,
bias=True)
                (value): Linear(in_features=768, out_features=768,
bias=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
              (output): BlipTextSelfOutput(
```

```
                (dense): Linear(in_features=768, out_features=768,
bias=True)
                (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
            )
            (crossattention): BlipTextAttention(
              (self): BlipTextSelfAttention(
                (query): Linear(in_features=768, out_features=768,
bias=True)
                (key): Linear(in_features=768, out_features=768,
bias=True)
                (value): Linear(in_features=768, out_features=768,
bias=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
              (output): BlipTextSelfOutput(
                (dense): Linear(in_features=768, out_features=768,
bias=True)
                (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
            )
            (intermediate): BlipTextIntermediate(
              (dense): Linear(in_features=768, out_features=3072,
bias=True)
              (intermediate_act_fn): GELUActivation()
            )
            (output): BlipTextOutput(
              (dense): Linear(in_features=3072, out_features=768,
bias=True)
              (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
              (dropout): Dropout(p=0.0, inplace=False)
            )
          )
        )
      )
    )
    (cls): BlipTextOnlyMLMHead(
      (predictions): BlipTextLMPredictionHead(
        (transform): BlipTextPredictionHeadTransform(
          (dense): Linear(in_features=768, out_features=768,
bias=True)
          (transform_act_fn): GELUActivation()
          (LayerNorm): LayerNorm((768,), eps=1e-12,
elementwise_affine=True)
```

```
        )
        (decoder): Linear(in_features=768, out_features=30524,
bias=True)
      )
    )
  )
)

uploaded_test_image2 = files.upload()
uploaded_test_image_path2 = list(uploaded_test_image2.keys())[0]
img2 = cv2.imread(uploaded_test_image_path2)
img_rgb2 = cv2.cvtColor(img2, cv2.COLOR_BGR2RGB)

inputs2 = processor(images=img_rgb2, return_tensors="pt")

with torch.inference_mode():
    out_ids2 = model.generate(
        **inputs2,
        max_new_tokens=120,
        num_beams=5,
        min_new_tokens=30,
        repetition_penalty=1.2,
    )

caption2 = processor.tokenizer.decode(out_ids2[0],
skip_special_tokens=True)
caption2 = caption2.replace("_", " ").strip()   # in case you see _
characters
print(caption2)
cv2_imshow(img2)
```

```
<IPython.core.display.HTML object>

Saving still-life-messy-office-desk.jpg to still-life-messy-office-
desk.jpg
a green table topped with a plate of food and a red telephone, a cup
of coffee, a plant, and a pair of scissors and a phone
```